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(54) **VALVE ASSEMBLY AND METHOD**

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F02M 26/38 (2016.01)
F02M 26/70 (2016.01)
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(52) **U.S. Cl.**

CPC **F16K 1/222** (2013.01); **F02M 26/38** (2016.02); **F02M 26/70** (2016.02); **F16K 1/223** (2013.01); **F16K 27/0218** (2013.01)

(58) **Field of Classification Search**

CPC F16K 1/222; F16K 1/223; F16K 27/0218; F02M 26/38; F02M 26/70
USPC 251/305–308; 123/336–337
See application file for complete search history.

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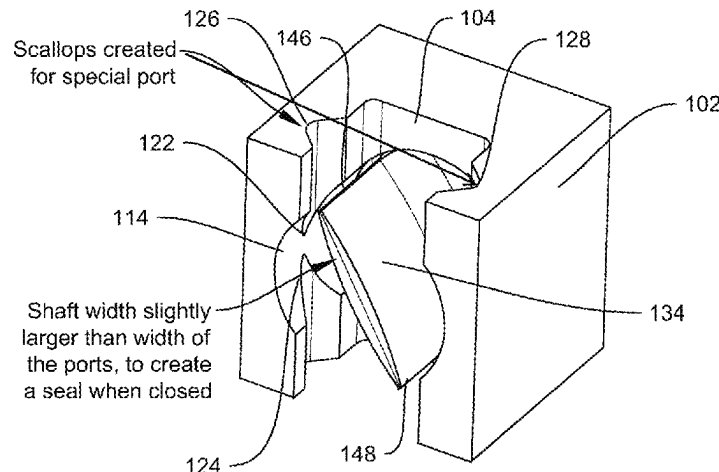
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(57) **ABSTRACT**

Embodiments of the present disclosure present a valve assembly that includes a valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first bearing surface concentric with the longitudinal axis and a radially spaced apart second bearing surface concentric with the longitudinal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface. The valve assembly further includes a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore.

20 Claims, 11 Drawing Sheets



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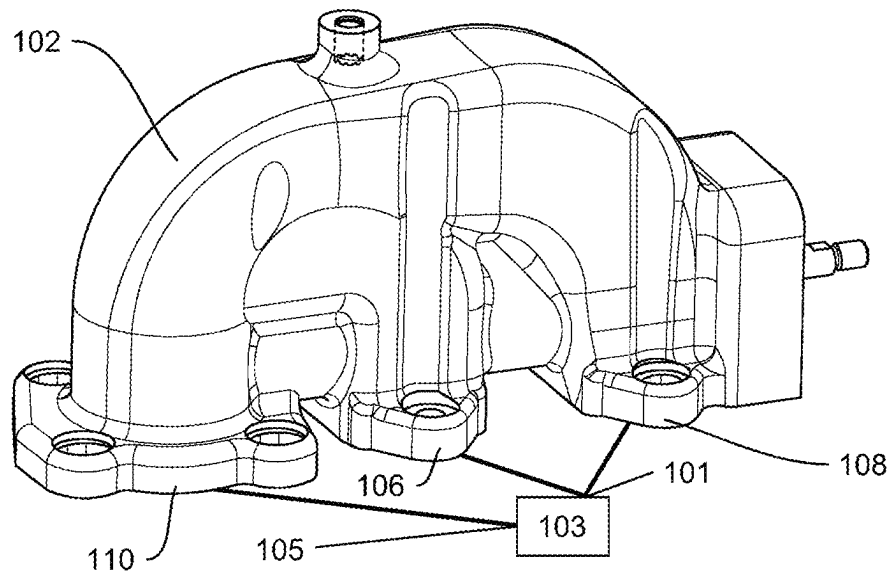


FIG. 1

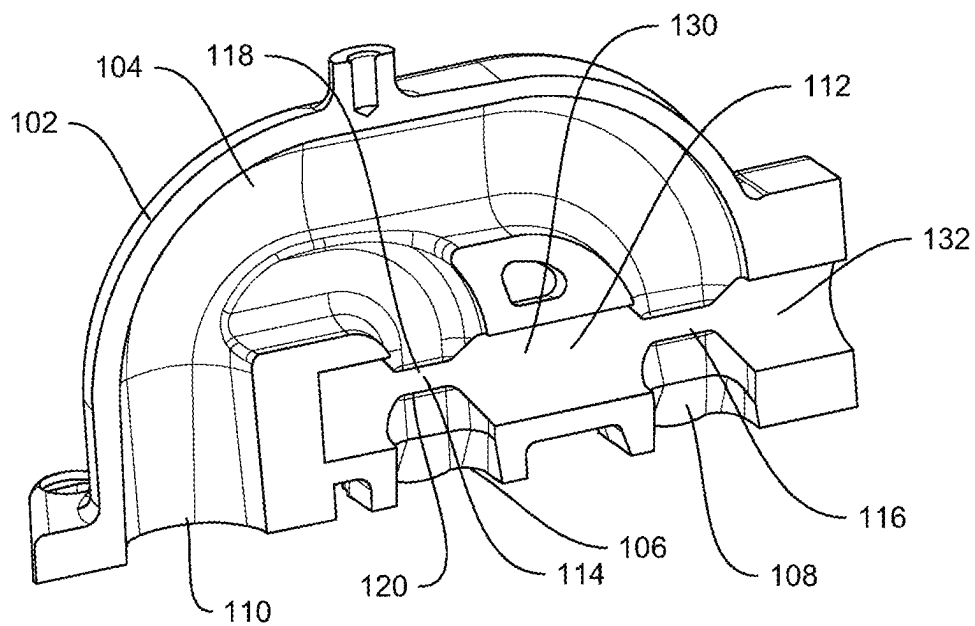


FIG. 2

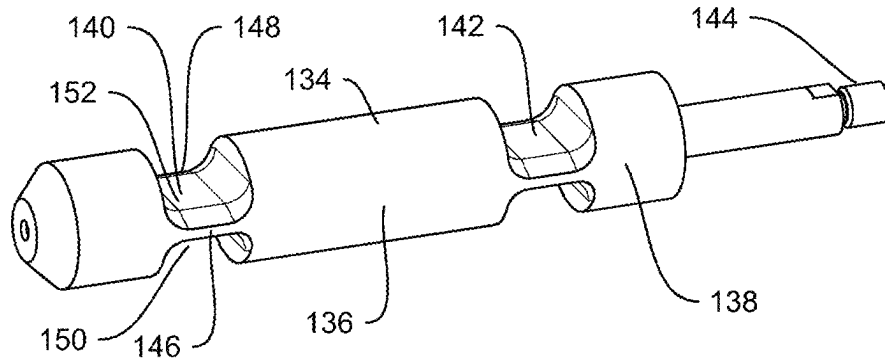


FIG. 3

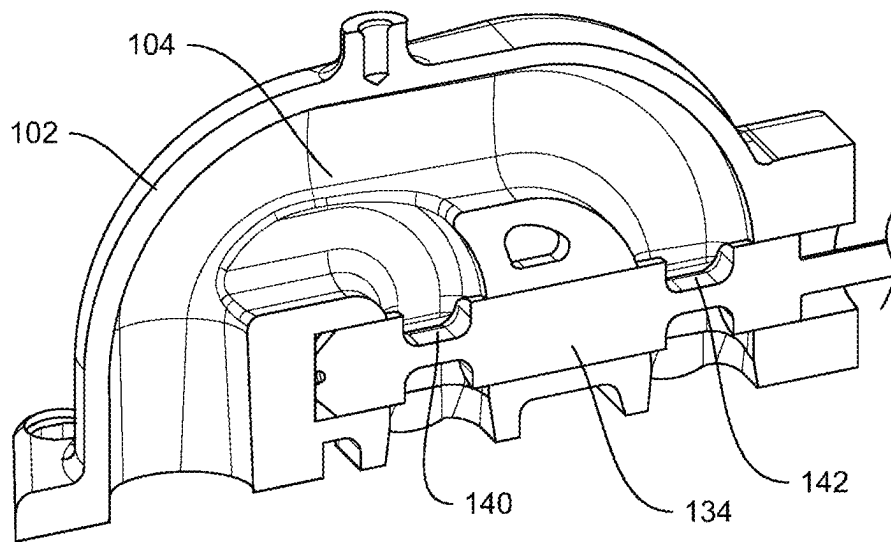


FIG. 4

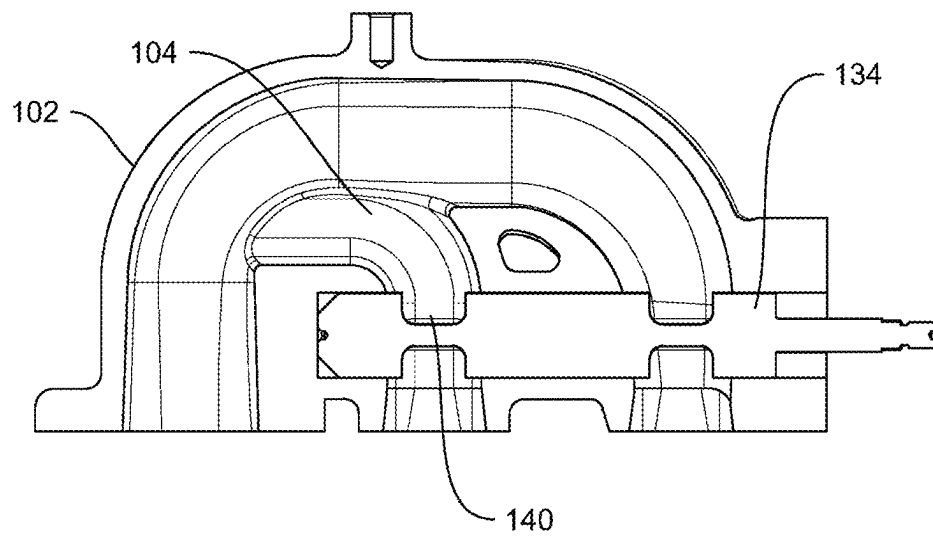


FIG. 5

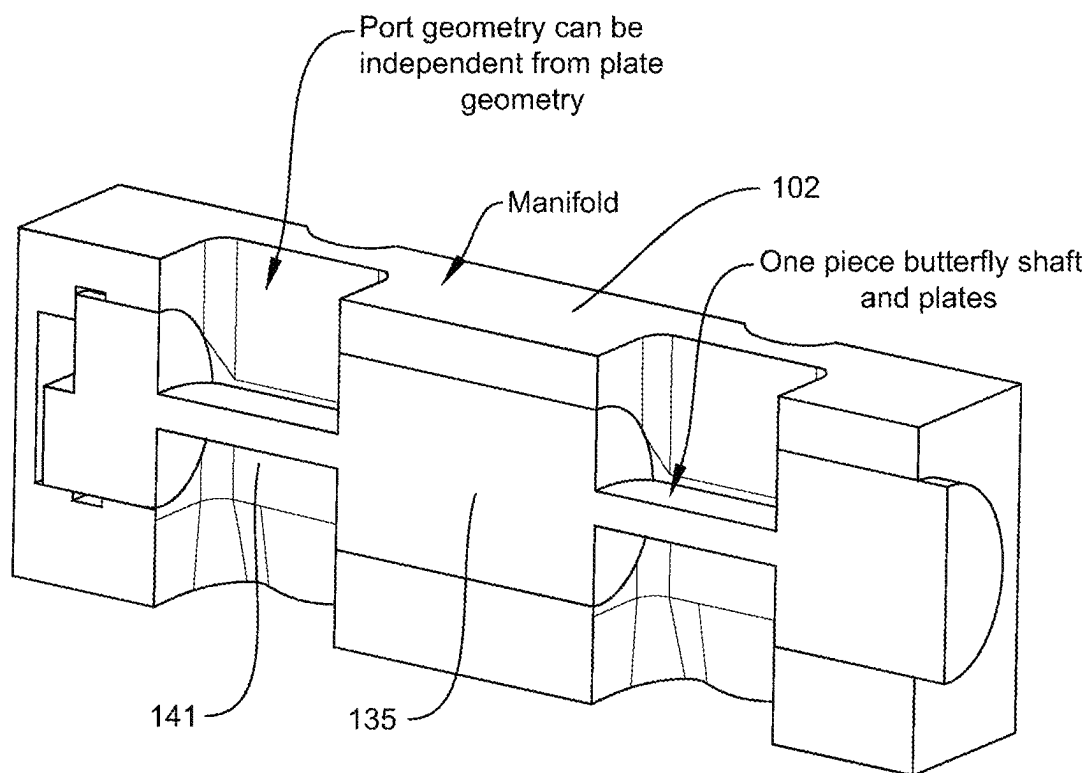


FIG. 6

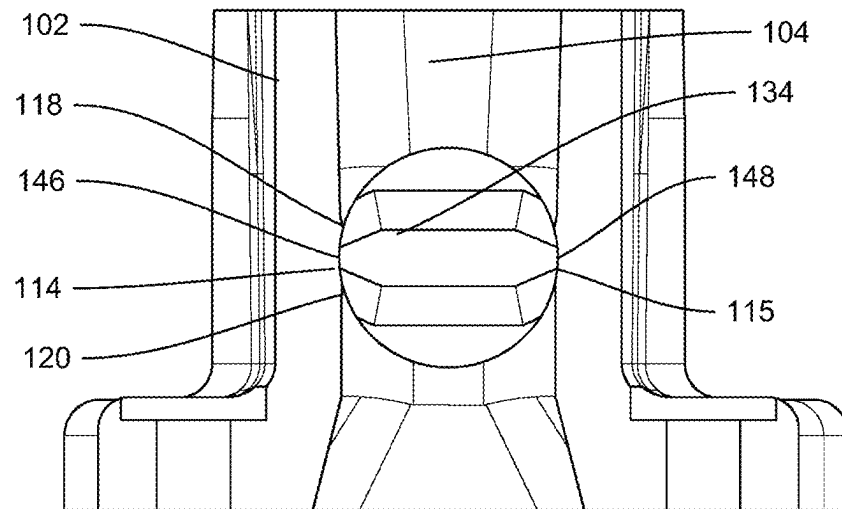


FIG. 7

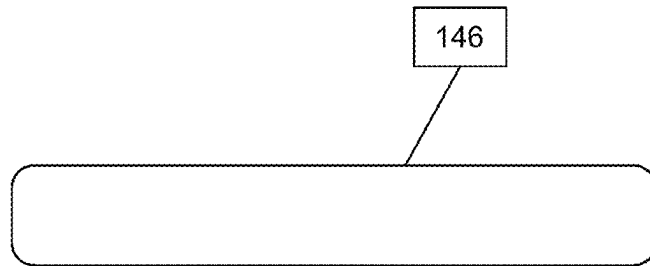


FIG. 8

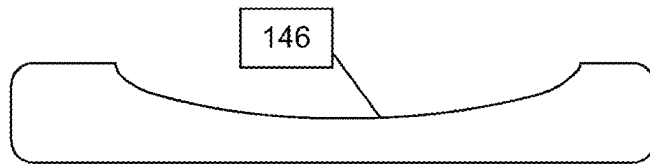


FIG. 9

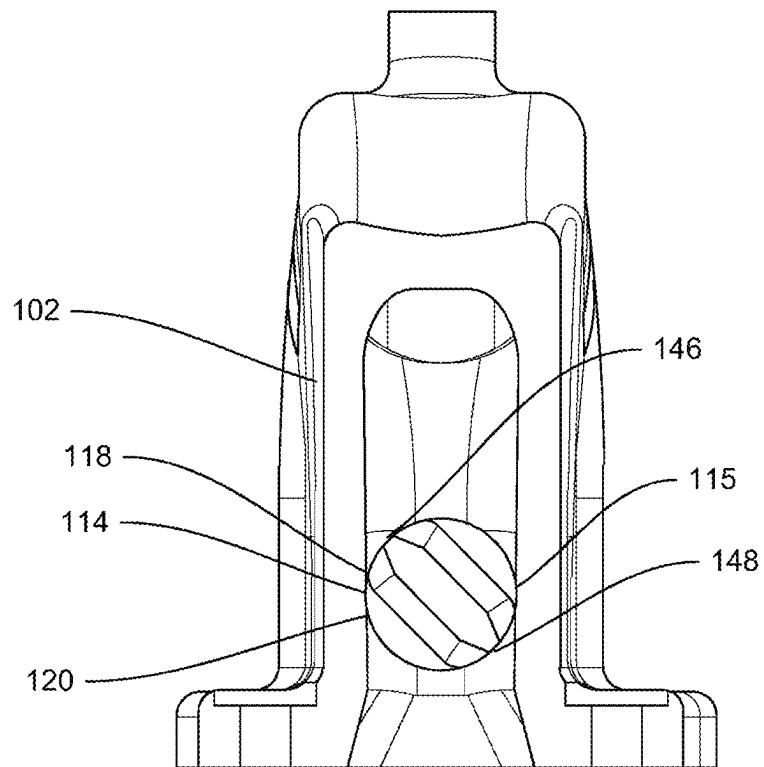
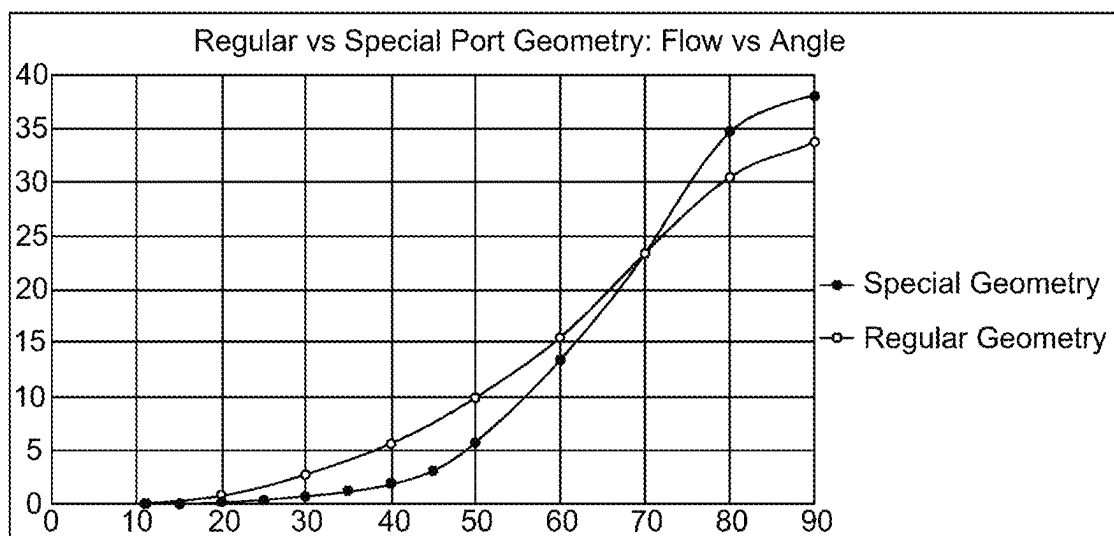
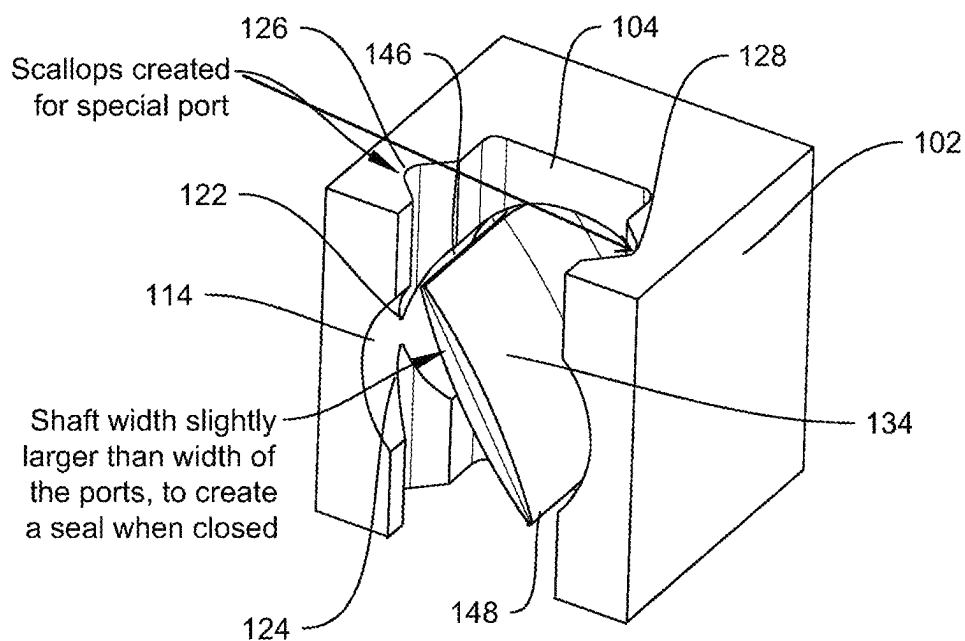


FIG. 10



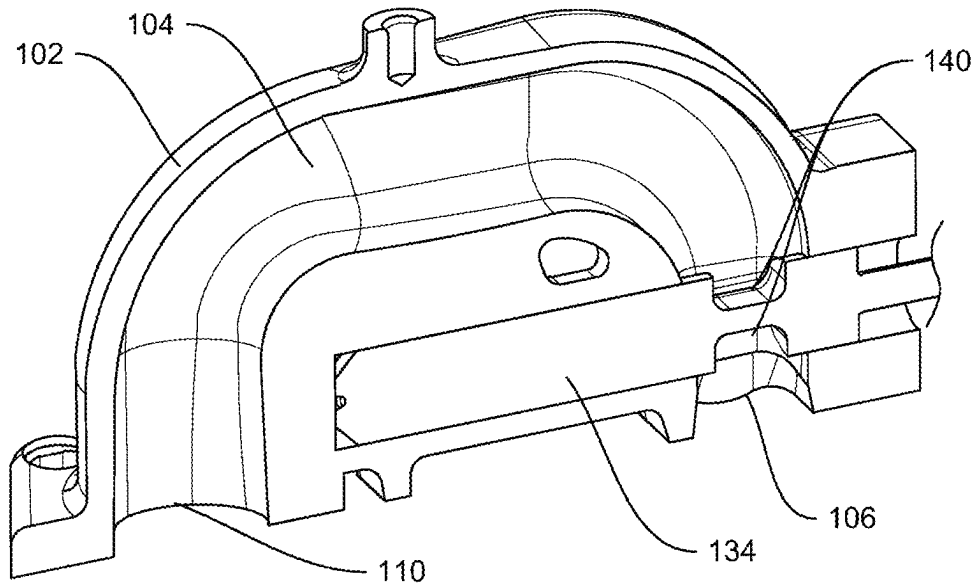


FIG. 13

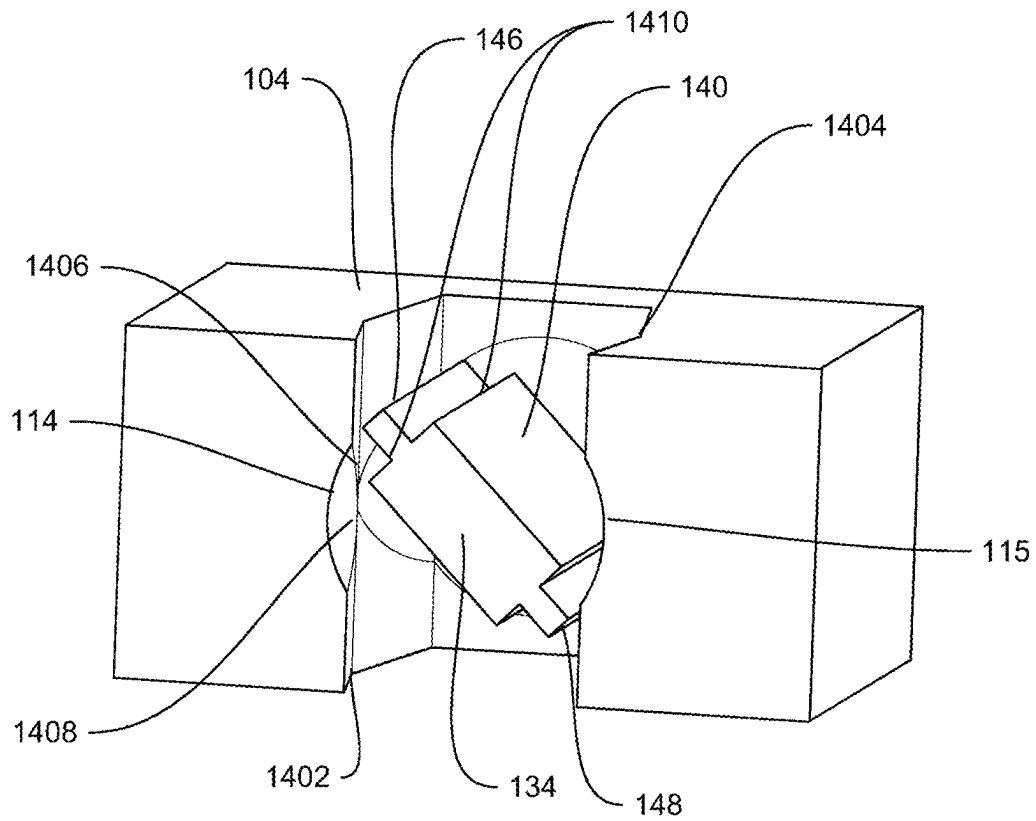


FIG. 14

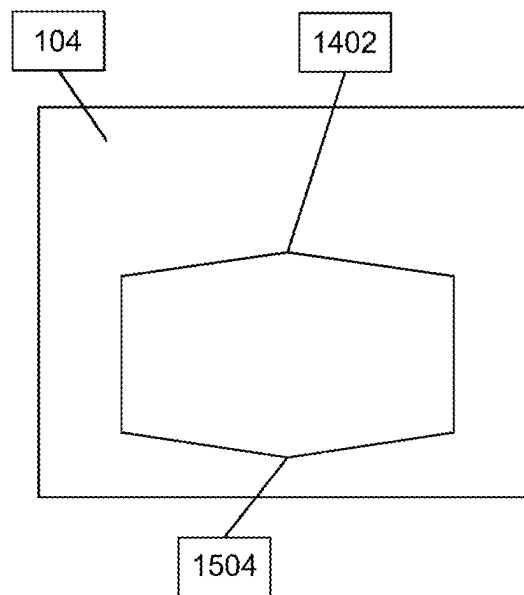


FIG. 15

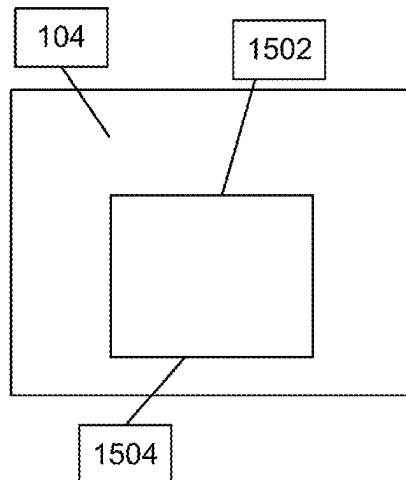


FIG. 16



FIG. 17

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VALVE ASSEMBLY AND METHOD

BACKGROUND

Field of the Disclosure

The present disclosure relates to valves and particularly to flow control including gas flow control and more particularly to butterfly type valves.

Description of Related Art

A valve is a device that is able to regulate or control the flow of a gas, liquid and/or particulates by opening and closing a channel or passageway. When a valve is in the open position, fluid is typically able to flow in a direction from high pressure to low pressure. When a valve is in the closed position, fluid flow is typically obstructed from flowing.

SUMMARY OF THE DISCLOSURE

In view of the foregoing, it is an object of the present disclosure to provide a method and apparatus.

A first exemplary embodiment of the present disclosure provides a valve assembly. The valve assembly includes a valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first annular bearing surface concentric with the longitudinal axis and a radially spaced apart second annular bearing surface concentric with the longitudinal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface. The valve assembly further includes a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore, the shaft valve having a first peripheral contact surface configured to engage the first bearing surface, a second peripheral contact surface configured to engage the second bearing surface and a vane radially intermediate the first peripheral contact surface and the second peripheral contact surface, wherein the first and the second peripheral contact surfaces have a given diameter and the vane encompasses the given diameter and wherein rotation of the shaft valve relative to the valve body selectively permits flow through the gas passage bore.

A second exemplary embodiment of the present disclosure provides an exhaust gas recirculation valve. The exhaust gas recirculation valve includes a valve body having a gas passage bore extending between at least one inlet and an at least one outlet, wherein the at least one inlet is configured to receive an exhaust gas from an exhaust manifold of an internal combustion engine and the at least one outlet is configured to pass exhaust to an inlet manifold of the internal combustion engine, the gas passage bore including a first inlet gas passage extending from the at least one inlet, the valve body having a valving bore intersecting the first inlet gas passage, the valving bore defining a first circular bearing surface and a second annular bearing surface. The exhaust gas recirculation valve further includes a shaft valve slideably received within the valving bore, the shaft valve having a first contact surface for rotatably engaging the first circular bearing surface and a second contact surface for rotatably engaging the second circular bearing surface, and a vane, wherein the first contact surface, the second contact surface and the vane have a common diameter.

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A third exemplary embodiment of the present disclosure provides a method. The method includes providing a valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first annular bearing surface concentric with the longitudinal axis and a radially spaced apart second annular bearing surface concentric with the longitudinal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface. The method further includes providing a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore, the shaft valve having a first peripheral contact surface configured to engage the first bearing surface, a second peripheral contact surface configured to engage the second bearing surface and a vane radially intermediate the first peripheral contact surface and the second peripheral contact surface, wherein the first and the second peripheral contact surfaces have a given diameter and the vane encompasses the given diameter and wherein rotation of the shaft valve relative to the valve body selectively permits flow through the gas passage bore.

A fourth exemplary embodiment of the present disclosure provides a method. The method includes forming a valve body having a gas passage bore extending between at least one inlet and an at least one outlet, wherein the at least one inlet is configured to receive an exhaust gas from an exhaust manifold of an internal combustion engine and the at least one outlet is configured to pass exhaust to an inlet manifold of the internal combustion engine, the gas passage bore including a first inlet gas passage extending from the at least one inlet, the valve body having a valving bore intersecting the first inlet gas passage, the valving bore defining a first circular bearing surface and a second annular bearing surface. The method further includes forming a shaft valve slideably received within the valving bore, the shaft valve having a first contact surface for rotatably engaging the first circular bearing surface and a second contact surface for rotatably engaging the second circular bearing surface, and a vane, wherein the first contact surface, the second contact surface and the vane have a common diameter.

The following will describe embodiments of the present disclosure, but it should be appreciated that the present disclosure is not limited to the described embodiments and various modifications of the invention are possible without departing from the basic principles. The scope of the present disclosure is therefore to be determined solely by the appended claims.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is an exterior view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 2 is a cross-sectional view of an exemplary valve body suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 3 is a perspective view of an exemplary shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 4 is a cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 5 is another cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 6 is yet another cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 7 is a front cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 8 is a side elevational view of a seal face suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 9 is a side elevational view of a shaped seal face suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 10 is another front cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 11 is a perspective cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 12 is a graph illustrating flow rates through a gas passage vs. angular location of a shaft within the gas passage in accordance with exemplary embodiments of this disclosure.

FIG. 13 is another embodiment of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 14 is another perspective cross-sectional view of an exemplary valve body and shaft valve suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 15 is a top view of the shape of the gas passage intersection with the valving bore suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 16 is another top view of the shape of the gas passage intersection with the valving bore suitable for use in practicing exemplary embodiments of this disclosure.

FIG. 17 is an exemplary logic flow diagram in accordance with a method for practicing exemplary embodiments of this disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

Embodiments of the present disclosure can be used in exhaust gas recirculating valves (EGR) connected to exhaust manifolds **101** of a combustion engine **103** (shown in FIG. 1) to divert metered amounts of the exhaust gas to intake manifolds **105** for re-burn by the engine. The exhaust gases are mixed with fresh air/fuel mixtures resulting in a lowering of combustion temperature and a reduction in the formation of harmful compounds such as nitrous oxide. Embodiments of the present disclosure provide a valve body and a single piece shaft valve having a vane with butterfly plate flow geometry. The diameter of the single piece shaft valve is relatively uniform throughout its longitudinal axis including the vane portion of the shaft valve.

It is contemplated that embodiments of the present disclosure can be used in place of traditional butterfly valves. In this embodiment, the valve will include a body having a relatively large-diameter first bore there through for passage of a gas, and a second relatively small-diameter bore transverse to the first bore for supporting a rotatable shaft on which is mounted a valve plate (known in the art as a "butterfly") for controllably occluding the first bore in response to rotation of the shaft to control the flow of gas.

Embodiments of the valve of the present disclosure include a valve body and a shaft valve rotatably connected to the valve body.

Valve Body

Referring to FIGS. 1 and 2, shown is a valve body **102**, which includes a gas passage bore **104** extending between an inlet and an outlet. It is contemplated in some configurations, the gas passage bore **104** will include a plurality of inlets or a plurality of outlets. As seen in FIG. 2, the valve body **102** includes two gas inlet passages **106**, **108** and a combined gas outlet passage **110**. The two exhaust gas inlet passages **106**, **108** admit exhaust gases from an engine exhaust manifold **101**. The exhaust gas outlet passage **110** directs a metered flow of the exhaust gases toward an engine inlet manifold **105**.

The gas passage bore **104** can have any of a variety of cross sectional profiles (e.g., circular, oval, rectangular, triangular, etc), wherein the profile is constant or varies along a length of the bore.

The valve body **102** also includes a cylindrical valving bore **112** extending along a longitudinal axis and intersecting the gas passage bore **104**. As seen in FIG. 2, the valving bore **112** extends through both inlet passages **106**, **108** of the gas passage bore **104**.

Each intersection of the valving bore **112** and the gas passage **104** defines a corresponding flow path cross sectional area.

Thus, the gas passage bore **104** can have a circular, obround, oval, rectilinear or faceted periphery, either at the intersection with the valving bore **112**, as well as upstream or downstream of the valving bore **112**.

A common or overlapping portion of the gas passage bore **104** and the valving bore **112** defines a sealing band **114**, **116** on opposite sides of the valving bore **112**, wherein the sealing bands **114**, **116** extend parallel to the longitudinal axis. As the valving bore **112** has a circular cross section, the sealing bands **114**, **116** have a generally arcuate cross section taken perpendicular to the longitudinal axis. Thus, the sealing bands **114**, **116** have a circumferential dimension about the longitudinal axis (shown in FIGS. 7, 10, 11 and 14). The sealing bands **114**, **116** extend about the periphery of the valving bore **112** as well as extending parallel to the longitudinal axis. In the configuration having a plurality of inlet passages (shown in FIG. 2), each intersection of the gas inlet passage **106**, **108** and the valving bore **112** includes a pair of opposing sealing bands **114**, **116**.

As set forth below, the sealing bands **114**, **116** can be defined by parallel edges **118**, **120**, each parallel to the longitudinal axis. Alternatively, the sealing bands **114**, **116** can include cut outs or a shaped edge **122**, **124** (shown in FIG. 11) so that flow can be permitted or precluded (depending on the shaping) relative to an adjacent portion of the sealing band **114**, **116**. That is, the sealing band **114**, **116** can have varying circumferential dimension along the longitudinal axis. For example, one end of the sealing band **114**, **116** may have a circumferential dimension of 0.5 cm (or an arc of 3 degrees) and the other end of the sealing band may have a circumferential dimension of 0.25 cm (or an arc of 1.5 degrees). As set forth below, this shaping can provide for increased low flow control of the valve. Additionally, the curvature of sealing bands **114**, **116** provide a larger flow area than gas bore passage **104** allowing for increased flow over sealing bands **114**, **116**.

In conjunction with the shaping of the sealing band **114**, **116**, the valve body **102** can include fluting or channels **126**, **128** (shown in FIG. 11) that extend from the sealing band **114** along the gas passage **104**. The valve body **102** can also include triangular or V-shaped channels **1402**, **1404** (shown in FIG. 14). In these embodiments, the terminal edge of channels **126**, **128** or the terminal edge **1406**, **1408** of

V-shaped channels **1402**, **1404** are adjacent to the sealing bands **114**, **116** and do not terminate at the same rotational location of the valve shaft **140**. Rather, the terminal edge **1406**, **1408** is also V-shaped such that it sealing bands **114**, **116** terminate at different rotational locations of the valve shaft **140**. A cross sectional top view of the valve body **104** at FIG. **15** illustrates the V-shaped channels **1402**, **1404**.

In another embodiment, the valve body **102** can have a substantially flat surface **1502**, **1504** adjacent the sealing bands **114**, **116**, and does not include channels **126**, **128**. This embodiment is shown in FIG. **16**, which depicts another cross sectional top view of valve body **104**.

The valving bore **112** also defines at least two or more bearing surfaces **130**, **132** (shown in FIG. **2**). As part of the valving bore **112**, the bearing surfaces **130**, **132** have the same diameter as the valving bore **112**. The bearing surfaces **130**, **132** are interfacing surfaces for supporting rotation of the shaft valve **134** relative to the valve body **102**. The bearing surfaces **130**, **132** are cylindrical in shape and have a diameter to provide for operable rotation of the shaft valve **134** relative to the valve body **102**.

Shaft Valve

The shaft valve **134** (shown in FIG. **3** and FIG. **4**) is a cylindrical member having a given diameter sized to be slidably received within the valving bore **112** and rotatably retained within the valving bore **112**.

The shaft valve **134** includes at least a first peripheral contact surface **136** configured to engage the first bearing surface **130**, a second peripheral contact surface **138** configured to engage the second bearing surface **132**. As seen in the figures, the shaft valve **134** can include a third, or additional, peripheral contact surfaces for rotatably engaging corresponding surfaces of the valve body **102**.

The peripheral contact surfaces **136**, **138** define a maximum diameter of the shaft valve **134**. Except for the vanes **140**, **142**, it is contemplated that other peripheral sections of the shaft valve **134** can have a slightly reduced diameter to facilitate operably locating the shaft valve **134** within the valving bore **112** and reduce rotating friction between the shaft valve **134** and the valve body **102**. However, excluding vanes **140**, **142**, it should be appreciated that embodiments of the shaft valve **134** include shaft valve **134** having a relatively uniform diameter throughout its longitudinal axis.

The sizing of the diameter of the shaft valve **134** to the valving bore **112** (and hence bearing surfaces **130**, **132**) is selected in accordance with standard engineering practices to permit the shaft valve **134** to be slid into the valving bore **112** and permit rotation of the shaft valve **134** relative to the valve body **102**. The diameter of the shaft valve **134** relative to the valving bore **112** is selected to permit rotation and accommodate tolerances of normal manufacturing processes. It should be appreciated that the shaft valve **134** is sized to valving bore **112** such that except for gas allowed to flow over the sealing surfaces and peripheral contact surfaces **136**, **138** when the shaft valve **134** is in the open position allowing gas to flow through gas passage bore **104**, gas is substantially obstructed from flowing between the shaft valve **134** and valving bore **112**.

As seen in the figures, the shaft valve **134** can include a length of reduced diameter to form a shank **144** for engaging a drive or control mechanism for selectively imparting rotation of the shaft valve **134** relative to the valve body **102**. The shank **144** can include a faceted portion or a flat or flats, keyways or splines, or any other shape to cooperatively engage with the drive or control mechanism. The drive or control mechanism can be any of a variety of configurations

including an engine control module (ECM) and an electric, electromechanical or hydraulic motor.

The shaft valve **134** also includes a vane **142** longitudinally intermediate the first peripheral contact surface **136** and the second peripheral contact surface **138**. As seen in the figures, the shaft valve **134** can include a number of vanes **140**, **142** corresponding to the number of gas passage bores **104**.

The vanes **140**, **142** are configured to selectively engage the sealing bands **114**, **116** to preclude (maximally inhibit) flow through the respective gas passage **114**, **116**. That is, each edge of the vane **140**, **142** includes a seal face **146**, **148**, wherein the distance between the seal faces **146**, is the diameter of the shaft valve **134**.

The vanes **140**, **142** are generally defined by a pair of opposite recesses **150**, **152** in the shaft valve **134**. Thus, by rotation of the shaft valve **134** relative to the valve body **102**, the vane **140** moves from a sealed orientation with each seal faces **146**, **148** engaging a corresponding sealing band **114**, **115** of the valve body **102** to a fully open position parallel to the flow.

The sealing bands **114**, **116** can have any of a variety of configurations and dimensions. For example, by forming the sealing bands **114**, **116** to have a circumferential dimension that is greater, such as by 1.1, 1.4, 1.6 or more times a dimension of the seal face **146**, **148** of the vane **140**, **142**.

The recesses **150**, **152** forming the vane **140** can be configured to provide the vane **140** with a constant thickness across the diameter of the shaft valve **134** or a varying thickness across the shaft valve **134**. As seen in FIGS. **7**, **10**, and **14**, the vane **140** has a first thickness encompassing the longitudinal axis and a reduced or lesser thickness at the seal face **146**, **148**.

Referring to FIGS. **7** and **10**, the seal face **146**, **148** is defined by generally parallel edges extending parallel to the longitudinal axis. That is, the seal face **146**, **148** has a generally rectangular shape (though having a curvature corresponding to the cylindrical periphery of the shaft valve **134**) (shown in FIGS. **8** and **9**). However, as shown in the figures, it is possible to shape the seal face **146**, **148** so that the seal face **146**, **148** has a varied circumferential dimension wherein flow would be permitted along some of the vane **140** while remaining precluded from a different portion of the vane **140** as the vane **140** rotates relative to the valve body **102**.

Similarly, the sealing band **114**, **115** and corresponding seal face **146**, **148** can be differently sized for a given vane **140**. This further allows for customized flow control as for a predestined rotation of the shaft valve **134**, gas can flow about one edge of the vane **140**, while the remaining edge of the vane **140** precludes flow. Embodiments include the valve shaft **134** having a substantially flat sealing face **146**, **148** as shown in FIG. **14**. In the embodiment depicted in FIG. **14**, vane **140** includes steps **1410**, which allow for vane **140** to be formed through less costly manufacturing means. In this embodiment the shaft width (i.e., distance between sealing face **146** and sealing face **148** is slightly larger than the distance between sealing bands **114**, **116** in order to create a sealed interface between the sealing faces **146**, **148** and the sealing bands **114**, **115** to prevent a flow of gas through the passage when the shaft valve **134** is in the closed position.

By selecting the relative sizing of the sealing bands **114**, **115** of the valve body **102** and the seal face **146**, **148** of the vane **140**, the shaft valve **134** can permit slight rotation of the vane **140** relative the valve body **102** without permitting flow. That is, the sensitivity of the shaft valve **134** to a zero or null position relative to the valve body **102** can be

reduced. Similarly, the sensitivity of the shaft valve **134** through low angle rotations relative to the valve body **102** can be reduced. As seen in FIG. **11**, approximately 5% of the flow can be controlled through approximately 50° of rotation. This is illustrated in the graph shown in FIG. **12**, which

depicts the overall percentage flow rate on the y-axis versus rotational degree of the shaft valve **134** on the x-axis. The sizing of the sealing band **114**, **115** and seal face **146**, **148** provide an overlapping interface that allows for the elimination of a conventional end stop point, as well as the elimination of valve shut off point shift due to thermal expansion dimensional changes at operating temperatures up to 720° C. Indeed, embodiments include the absence of a mechanical or physical stop or obstruction within or adjacent to the sealing bands **114**, **115** that prevent rotation of valve shaft **134**. In other words, embodiments include valve shaft **134** being operable to rotate in multiple directions and to multiple degrees greater than 360 degrees in either direction (e.g., clockwise or counterclockwise).

Referring to the figures, the shaft valve **134** can be used to set a maximum flow rate of the system. That is, by sizing the vane **140** to have a longitudinal dimension that is, for example 90%, 80% or 50% of the dimension of the gas passage bore **104** along the longitudinal axis, the shaft valve **134** can define a maximum flow path for the system. This allows the valve body **102**, and corresponding gas passage bore **104** to be sized to accommodate a relatively large system requiring a relatively high flow rate, wherein the longitudinal dimension of the vane **140** matches the longitudinal dimension of the associated gas passage bore **104**. However, since the shaft valve **134** can be readily replaced by merely sliding out a first shaft valve **135** having a vane **141** (shown in FIG. **6**) and sliding in a second shaft valve **134**, the second shaft valve **134** can have a vane **140** defined by a smaller longitudinal dimension, thereby restricting the maximum available flow rate of the valve. As shown in FIG. **6**, the longitudinal dimension of vane **141** is coextensive with the longitudinal dimension of gas passage bore **104**. Conversely, as shown in FIG. **5**, the longitudinal dimension of vane **140** is smaller than the longitudinal dimension of gas passage bore **104**. This allows a single size valve body **102** to be used in multiple operating environments.

As the shaft valve **134** can be used to limit the effective cross sectional area of the gas passage bore **104**, customization of the system is achieved by merely exchanging shaft valves **134** of different vane **140** configurations. As seen in the figures the longitudinal dimension of the vane **140** can be less than the longitudinal dimension of the gas passage bore **104** at the interface with the valving bore **112**. Thus, the shaft valve **134** effectively determines the available cross sectional flow area and hence flow rate.

It should be appreciated that while embodiments of the present valve assembly include a valve body **102** having two gas inlet passages **106**, **108** and a single gas outlet passage **110** (shown in FIG. **2**), embodiments of valve body **102** include a single gas inlet passage **106** and a single gas outlet passage **110** (shown in FIG. **13**). In this embodiment, shaft valve **134** includes a single vane **140** configured to selectively engage the sealing bands **114**, **115** to preclude flow through the gas passage **104**.

The presented embodiments offer significant advantages. These advantages include a reduced number of parts which lowers manufacturing costs as well as reduces the necessary steps for assembly. Further, as set forth above, as the shaft valve **134** can be replaced by merely sliding the old shaft valve **134** out of the valve body **102** and sliding the new shaft valve **134** into the valve body **102**, both component

cost and assembly cost are reduced. Also, as the number of parts is reduced, the necessary accounting for accumulated tolerances is reduced, thereby increasing manufacturing pass rates.

The ability of the present system to confront an area of the sealing band **114**, **115** with an area of the seal face **146**, **148**, rather than line contacts of the prior designs, the present system can reduce leakage from approximately 30 Kg/hr to less than 10 Kg/hr and in select configurations less than 1 Kg/hr. This reduced leakage is believed to be necessary to meet more stringent emissions standards being adopted globally.

Because the contact surfaces of the shaft valve **134** have the same diameter as the vane **140**, the load on the bearing surfaces is less than in a traditional butterfly valve. In certain configurations, the bearing load is decreased by a factor of 4. As there is reduced bearing load, alternative (and cheaper) materials can be used for the valve body **102** as well as the shaft valve **134**. Thus, less exotic materials for the valve body **102** and shaft valve **134** are required.

The shaped sealing band **114**, **115** in conjunction with the channels **126**, **128** in the valve body **102** provides for the configuration of the port, such as the "V" shape shown in FIG. **11**, which improves a shearing action of accumulated combustion materials, such as soot and carbon. The improved shearing results from the seal face **146**, **148** of the vane **140** progressively coming into engagement with the sealing band **114**, **115**, rather than the entire seal face **146**, **148** engaging the entire sealing band **114**, **115** at one time. By progressively engaging, a lower torque control motor or drive can be used. The lower torque motor is less costly and typically adds less weight to the vehicle in which the present valve is installed. It should be appreciated that while FIGS. **11**, **14**, and **15** depict "V" shaped channels **126**, **128**, embodiments of channels **126**, **128** can be shaped to include rectangular (shown in FIG. **16**), circular, oval (shown in FIG. **9** as sealing face **146**) or multiple "V" shaped channels in order to allow for different types of flow rate control through gas bore passage **104**.

As set forth above, the shaped sealing band and/or seal face which provide for the port configuration options such as "V" or other shapes change flow output versus angular rotation of the shaft valve **134**. This allows customization of the valve output flow for improved sensitivity at lower opening angles of the valve improving valve response. Additionally, the "V" shaped channels **126**, **128** not only allows for finer adjustments in the amount of flow through gas bore passage **104**, but channels **126**, **128** allow for finer adjustments in the velocity of flow through gas bore passage **104**. That is, rotation of valve shaft **134** allowing flow through only channels **126**, **128** will have a higher flow velocity when compared to the flow velocity when the sealing face **146**, **148** is no longer in contact with any portion of the sealing band **114**, **115**.

Also, the use of the shaft valve **134** provides flexibility to adjust maximum flow output by adjusting the machine depth or width of the recesses or cut outs in the shaft valve **134** as well as the number of cut outs without changing the package size or port configuration in the valve body **102**.

Reference is now made to FIG. **17**, which presents an exemplary method or process for providing embodiments of the present disclosure. The process begins at block **1702**, which states (a) providing valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first bearing surface concentric with the longitudinal axis and a radially spaced apart second bearing surface concentric with the longitudi-

nal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface; and (b) providing a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore, the shaft valve having a first peripheral contact surface configured to engage the first bearing surface, a second peripheral contact surface configured to engage the second bearing surface and a vane radially intermediate the first peripheral contact surface and the second peripheral contact surface, wherein the first and the second peripheral contact surfaces have a given diameter and the vane encompasses the given diameter and wherein rotation of the shaft valve relative to the valve body selectively permits flow through the gas passage bore. Following block 1702, block 1704 indicates wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal axis, each sealing band configured to engage a corresponding longitudinal edge of the vane.

Some of the other non-limiting embodiments include block 1706, which specifies wherein the sealing bands have one of (i) a V-shaped edge adjacent a channel defined in the valve body operable to allow a flow to pass therethrough, (ii) a rectangular shaped edge, and (iii) an oval shaped edge. Next, block 1708 states wherein the first peripheral contact surface configured to engage the first bearing surface and the second peripheral contact surface configured to engage the second bearing surface are operable to contact accumulated combustion products disposed on an edge of the sealing bands. Block 1710 indicates wherein the first peripheral contact surface configured to engage the first bearing surface and wherein the shaft valve is operable to rotate unobstructed 360 degrees within the valving bore.

The invention has been described in detail with particular reference to a presently preferred embodiment, but it will be understood that variations and modifications can be effected within the spirit and scope of the invention. The presently disclosed embodiments are therefore considered in all respects to be illustrative and not restrictive. The scope of the invention is indicated by the appended claims, and all changes that come within the meaning and range of equivalents thereof are intended to be embraced therein.

The invention claimed is:

1. A method comprising:

- (a) providing valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first bearing surface concentric with the longitudinal axis and a radially spaced apart second bearing surface concentric with the longitudinal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface; and
- (b) providing a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore, the shaft valve having a first peripheral contact surface configured to engage the first bearing surface, a second peripheral contact surface configured to engage the second bearing surface and a vane radially intermediate the first peripheral contact surface and the second peripheral contact surface, wherein the first and the second peripheral contact surfaces have a given diameter and the vane encompasses the given diameter and wherein rotation of the shaft valve relative to the valve body selectively permits flow through the gas passage bore, wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal

axis, each sealing band configured to engage a corresponding longitudinal edge of the vane, and wherein the valve body includes a first and a second opposing channel each extending perpendicular to the longitudinal axis through one of the pair of sealing bands.

2. The method of claim 1, wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal axis, each sealing band configured to engage a corresponding longitudinal edge of the vane.

3. The method of claim 2, wherein the sealing bands have one of (i) a V-shaped edge adjacent a channel defined in the valve body operable to allow a flow to pass therethrough, (ii) a rectangular shaped edge, and (iii) an oval shaped edge.

4. The method of claim 2, wherein the first peripheral contact surface configured to engage the first bearing surface and the second peripheral contact surface configured to engage the second bearing surface are operable to contact accumulated combustion products disposed on an edge of the sealing bands.

5. The method of claim 1, wherein the shaft valve is operable to rotate unobstructed 360 degrees within the valving bore.

6. A valve assembly comprising:

- (a) a valve body having a gas passage bore, a valving bore extending along a longitudinal axis and intersecting the gas passage bore, a first annular bearing surface concentric with the longitudinal axis and a radially spaced apart second annular bearing surface concentric with the longitudinal axis, wherein an interface of the gas passage bore and the valving bore defines a flow port radially intermediate the first bearing surface and the second bearing surface; and
- (b) a shaft valve extending along the longitudinal axis and rotatably mounted in the valving bore, the shaft valve having a first peripheral contact surface configured to engage the first bearing surface, a second peripheral contact surface configured to engage the second bearing surface and a vane radially intermediate the first peripheral contact surface and the second peripheral contact surface, wherein the first and the second peripheral contact surfaces have a given diameter and the vane encompasses the given diameter and wherein rotation of the shaft valve relative to the valve body selectively permits flow through the gas passage bore, wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal axis, each sealing band configured to engage a corresponding longitudinal edge of the vane, and wherein the valve body includes a first and a second opposing channel each extending perpendicular to the longitudinal axis through one of the pair of sealing bands.

7. The valve assembly of claim 6, wherein the vane is defined by opposing recesses in the shaft valve.

8. The valve assembly of claim 6, wherein the vane is defined by opposite major surfaces, and at least a portion of the major surfaces are non-parallel.

9. The valve assembly of claim 6, wherein a thickness of the vane varies along a radius extending from the longitudinal axis.

10. The valve assembly of claim 6, wherein the vane has a cross section substantially perpendicular to the longitudinal axis defining non-parallel major surfaces.

11. The valve assembly of claim 6, wherein the vane has a cross section substantially perpendicular to the longitudinal axis defining a first thickness at a first radial location along the longitudinal axis and a different second thickness at a second radial location along the longitudinal axis.

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12. The valve assembly of claim 6, wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal axis, each sealing band configured to engage a corresponding longitudinal edge of the vane.

13. The valve assembly of claim 6, wherein the sealing bands have one of (i) a V-shaped edge adjacent a channel defined in the valve body operable to allow a flow to pass therethrough, (ii) a rectangular shaped edge, and (iii) an oval shaped edge.

14. The valve assembly of claim 6, wherein the first peripheral contact surface configured to engage the first bearing surface and the second peripheral contact surface configured to engage the second bearing surface are operable to contact accumulated combustion products disposed on an edge of the sealing bands.

15. The valve assembly of claim 6, wherein the shaft valve is operable to rotate unobstructed 360 degrees within the valving bore.

16. The valve assembly of claim 6, wherein the sealing bands have a sealing diameter that is greater than a length of the longitudinal edge of the vane, and wherein the shaft valve and the valve body are operable to prevent a flow of gas between the shaft valve and the valve body when the longitudinal edge of the vane is in contact with the sealing bands.

17. The valve assembly of claim 6, wherein one of the valve body and vane has a shaped sealing surface that defines an opening cross sectional area that varies corresponding to an amount of rotation of the shaft valve operable to adjust flow rates through the gas passage bore relative to an angular position of the shaft valve.

18. An exhaust gas recirculation valve comprising:

- (a) a valve body having a gas passage bore extending between at least one inlet and an at least one outlet,

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wherein the at least one inlet is configured to receive an exhaust gas from an exhaust manifold of an internal combustion engine and the at least one outlet is configured to pass exhaust to an inlet manifold of the internal combustion engine, the gas passage bore including a first inlet gas passage extending from the at least one inlet, the valve body having a valving bore intersecting the first inlet gas passage, the valving bore defining a first circular bearing surface and a second annular bearing surface; and

- (b) a shaft valve slideably received within the valving bore, the shaft valve having a first contact surface for rotatably engaging the first circular bearing surface and a second contact surface for rotatably engaging the second circular bearing surface, and a vane, wherein the first contact surface, the second contact surface and the vane have a common diameter, wherein the valve body defines a pair of opposing sealing bands extending parallel to the longitudinal axis, each sealing band configured to engage a corresponding longitudinal edge of the vane, and wherein the valve body includes a first and a second opposing channel each extending perpendicular to the longitudinal axis through one of the pair of sealing bands, and wherein the sealing bands have a V-shaped edge adjacent a channel defined in the valve body operable to allow a flow to pass therethrough.

19. The exhaust gas recirculation valve according to claim 18, wherein the shaft valve is operable to rotate unobstructed by the valve body 360 degrees within the valving bore.

20. The exhaust gas recirculation valve of claim 18, wherein the vane is defined by opposing recesses in the shaft valve.

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